

V2I-BASED SMART TRAFFIC SIGNAL SYSTEM WITH ADAPTIVE TIMING AND EMERGENCY VEHICLE PREEMPTION

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BRAINSTORMING REPORT

IDENTIFIED PROBLEM

The existing traffic signal systems primarily operate using fixed-time schedules, which are unable to adapt to changing traffic conditions. As a result, vehicles experience unnecessary waiting even when roads are less congested, leading to increased traffic congestion, fuel consumption, travel time, and carbon emissions. Although some adaptive traffic control systems adjust signal timing based on vehicle density, they react only after congestion has already formed and do not predict the arrival of approaching vehicles. Furthermore, emergency vehicles such as ambulances, fire trucks, and police vehicles often face delays at intersections because existing systems lack an efficient mechanism to provide immediate signal priority. Many advanced traffic management solutions also depend on expensive communication technologies and infrastructure, making them difficult to implement on a large scale. These limitations highlight the need for a cost-effective, intelligent traffic management system capable of adaptive signal control, predictive traffic management, and emergency vehicle prioritization.

PROPOSED SOLUTION

1. Develop a V2I-Based Smart Traffic Signal System for intelligent traffic management.
2. Establish wireless communication between vehicles and traffic signals using nRF24L01 modules.
3. Use an ultrasonic sensor to measure real-time traffic density by detecting vehicle queue length.
4. Dynamically adjust green signal duration based on current traffic density.
5. Implement predictive green time by estimating the arrival time of approaching vehicles using their speed and approach information.
6. Detect emergency vehicles and automatically provide signal priority through emergency vehicle preemption.
7. Interrupt the normal traffic signal cycle during emergencies and restore normal operation after the emergency vehicle passes.
8. Reduce traffic congestion, vehicle waiting time, fuel consumption, and carbon emissions through intelligent signal control
9. Implement the system using STM32, nRF24L01, and embedded software for a low-cost and scalable solution
10. Provide a prototype that can be extended to multi-intersection traffic control and future smart city applications.

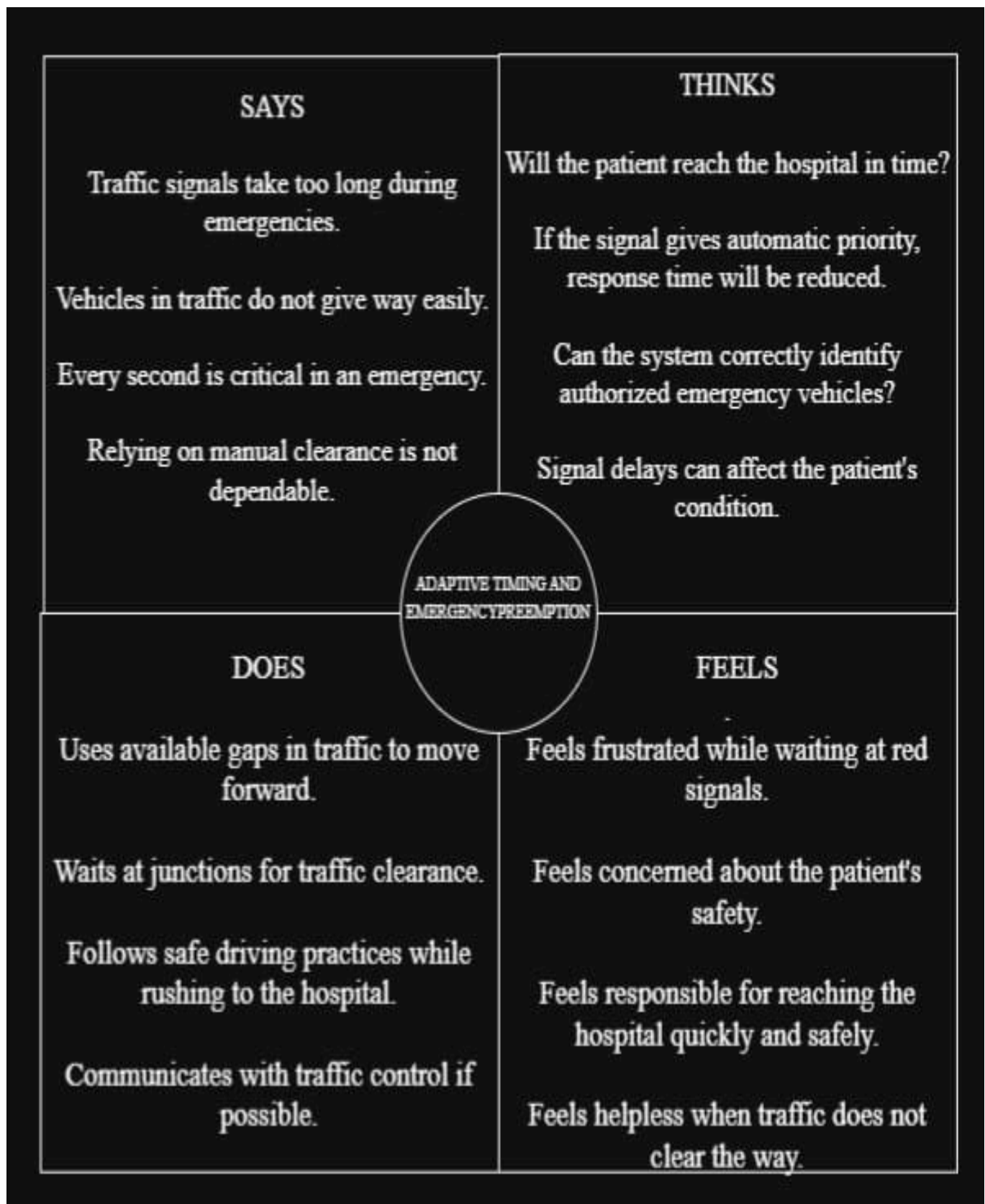
EXPECTED BENEFITS

The proposed system is expected to reduce traffic congestion, minimize vehicle waiting time, improve emergency response, reduce fuel consumption and carbon emissions, and enhance road safety. By integrating adaptive timing, predictive green time, and emergency vehicle preemption into a single low-cost embedded platform using STM32/Arduino and nRF24L01, the project provides a practical, scalable, and cost-effective solution for future intelligent transportation systems

IDEA PRIORITIZATION MATRIX



EMPATHY MAP



FUNCTIONAL REQUIREMENT

Vehicle Detection

The system shall detect vehicles approaching and waiting at the traffic signal.

Traffic Density Estimation

The system shall estimate the queue length/traffic density using vehicle detection sensors.

Adaptive Signal Timing

The system shall automatically increase or decrease the green-light duration based on the detected traffic density.

Emergency Vehicle Detection

The system shall receive an emergency message from an equipped emergency vehicle.

Emergency Vehicle Authentication

The system shall verify the emergency vehicle ID before granting priority.

Emergency Signal Preemption

When an authorized emergency vehicle is detected, the system shall interrupt the current signal cycle and provide green light in its direction.

Safe Signal Control

Other directions shall remain red while emergency priority is active to provide a clear path.

Normal Operation Recovery

After the emergency vehicle passes, the system shall automatically return to normal adaptive signal operation.

Wireless Communication

The system shall enable wireless communication between the emergency vehicle unit and traffic signal unit.

Status Display

The system shall display information such as traffic density, green time, vehicle type, and emergency status on an LCD/OLED.

RISK MATRIX

S.NO	RISK	IMPACT	SOLUTION
1	nRF24L01 communication failure	High	Check wiring, power supply and communication range
2	Emergency vehicle not detected	High	Use repeated emergency messages and proper authentication
3	Unauthorized emergency request	High	Verify the registered vehicle ID
4	Vehicle detection error	Medium	Use multiple ultrasonic sensors and proper sensor placement
5	Incorrect green timing	Medium	Set minimum and maximum green-time limits
6	Power failure	High	Provide backup battery/power supply
7	Signal conflict during emergency	Very High	Keep all other directions red before emergency green
8	Wireless interference	Medium	Select a suitable RF channel and use retransmission

GANTT CHART

